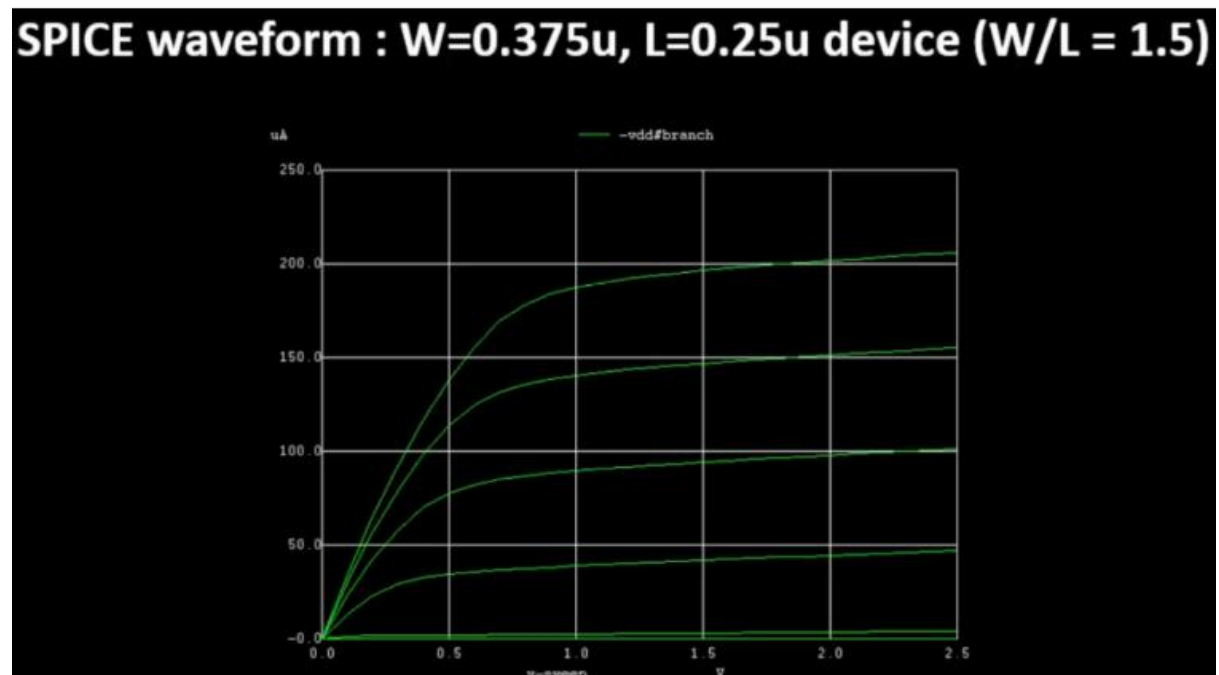
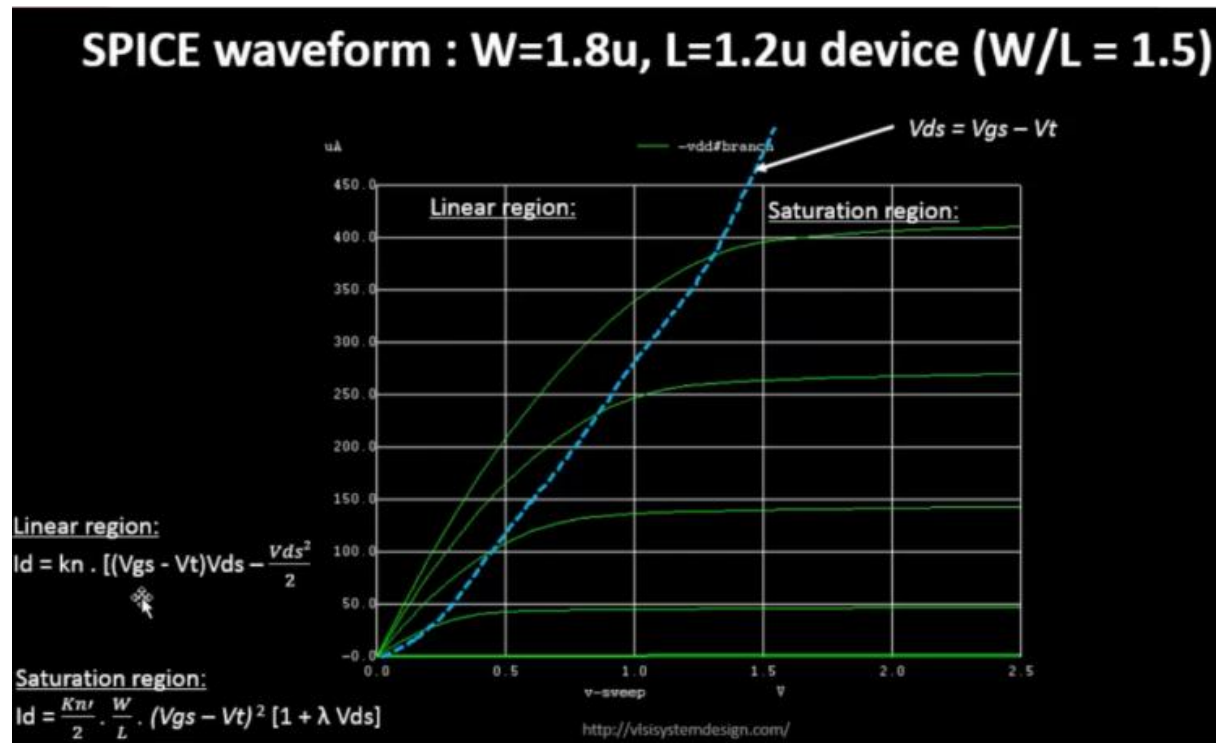
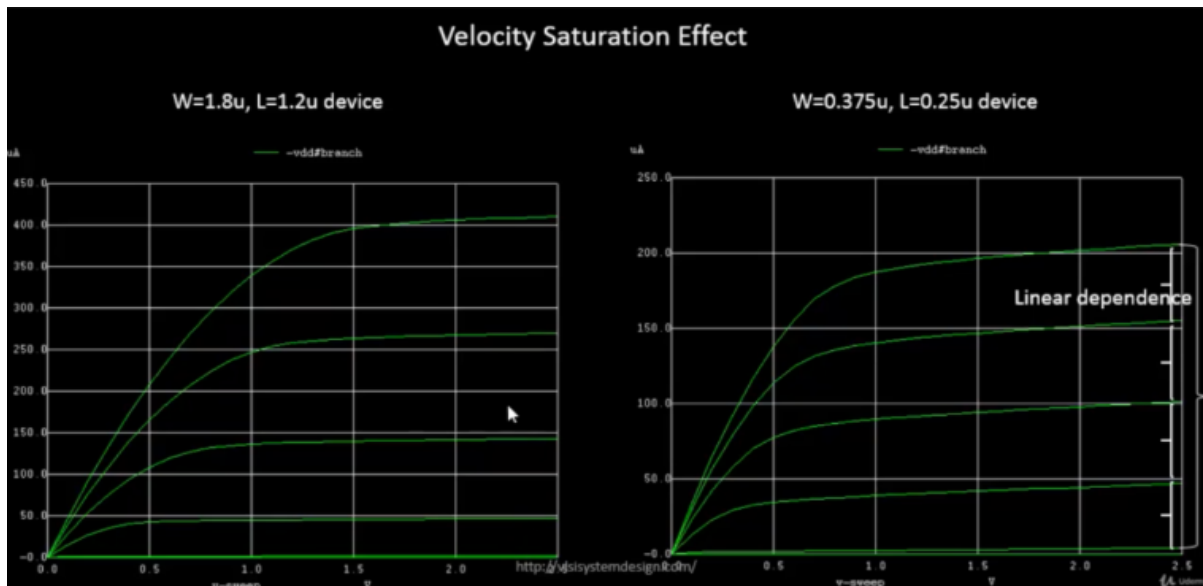


SPICE SIMULATION FOR LOWER NODES AND VELOCITY SATURATION EFFECT

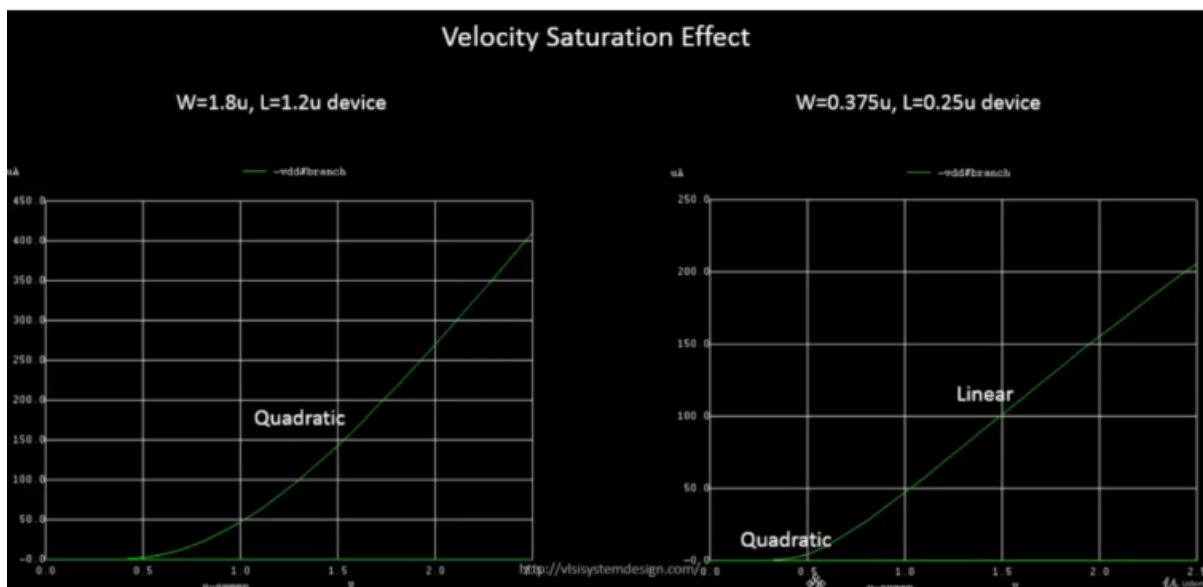
SPICE simulation for lower nodes





For **long-channel devices**, drain current shows a **quadratic dependence** on gate voltage.

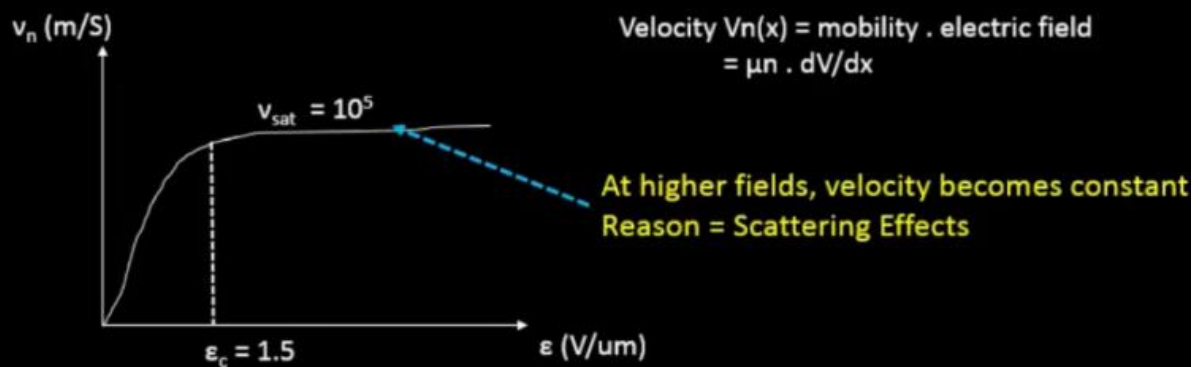
For **short-channel devices**, it is **quadratic at low gate voltage** but becomes **linear at higher voltages** due to **velocity saturation**.



At **lower electric fields**, carrier velocity increases **linearly** with the electric field.

At **higher electric fields**, velocity **saturates** and becomes **constant** due to **velocity saturation**.

Velocity Saturation Effect



$$v_n \text{ (m/S)} = \mu_n \cdot \epsilon / 1 + (\epsilon / \epsilon_c) \text{ for } \epsilon \leq \epsilon_c$$

$$= v_{sat} \text{ for } \epsilon \geq \epsilon_c$$

$$\epsilon_c = 2 v_{sat} / \mu_n$$

Velocity Saturation Effect

$$Q_i(x) = -C_{ox} ([V_{gs} - V(x)] - V_t)$$

$$\text{Velocity } V_n(x) = \text{mobility} \cdot \text{electric field}$$

$$= \mu_n \cdot dV/dx$$

Re-deriving drain current

using below boundary condition

$$I_d = -V_n(x) \cdot Q_i(x) \cdot W \rightarrow I_d = \frac{\mu_n \cdot C_{ox}}{1 + \left(\frac{V_{ds}}{\epsilon_c L}\right)} \cdot (W/L) [(V_{gs} - V_t)V_{ds} - V_{ds}^2/2]$$

$$v_n \text{ (m/S)} = \mu_n \cdot \epsilon / 1 + (\epsilon / \epsilon_c) \text{ for } \epsilon \leq \epsilon_c$$

$$= v_{sat} \text{ for } \epsilon \geq \epsilon_c$$

$$\epsilon_c = 2 v_{sat} / \mu_n$$

Velocity Saturation Effect

Let's come up with one model

Operation modes

Let's call $(V_{gs} - V_t) = V_{gt}$

Long Channel (>250nm)	Short Channel (<250nm)
Cutoff	Cutoff
Resistive	Resistive
	Velocity Saturation
Saturation	Saturation

$I_d = 0$, for $V_{gt} < 0$, cutoff mode

For all other modes, let's use below model

$$I_d = k_n \cdot [(V_{gt} \cdot V_{min}) - \frac{V_{min}^2}{2}] \cdot [1 + \lambda V_{ds}]$$

Where $V_{min} = \min(V_{gt}, V_{ds}, V_{dsat})$

Velocity Saturation Drain Current Model:

- **Technology Parameter:** Saturation voltage (V_{dsat}) – the voltage at which carrier velocity saturates, independent of V_{gs} or V_{ds} .
- V_{gt} is minimum $\rightarrow$ FET is in **saturation region** (both long and short channel).
- V_{ds} is minimum $\rightarrow$ FET is in **linear region** (both long and short channel).
- V_{dsat} is minimum $\rightarrow$ FET velocity **saturates** (**only for short channel**).

Velocity Saturation Effect

Let's come up with one model

Operation modes Let's call $(V_{gs} - V_t) = V_{gt}$

Long Channel (>250nm)	Short Channel (<250nm)
Cutoff	Cutoff
Resistive	Resistive
	Velocity Saturation
Saturation	Saturation

Id = 0, for $V_{gt} < 0$, cutoff mode

For all other modes, let's use below model

$$I_d = k_n \cdot \left[(V_{gt} \cdot V_{min}) - \frac{V_{min}^2}{2} \right] \cdot [1 + \lambda V_{ds}]$$

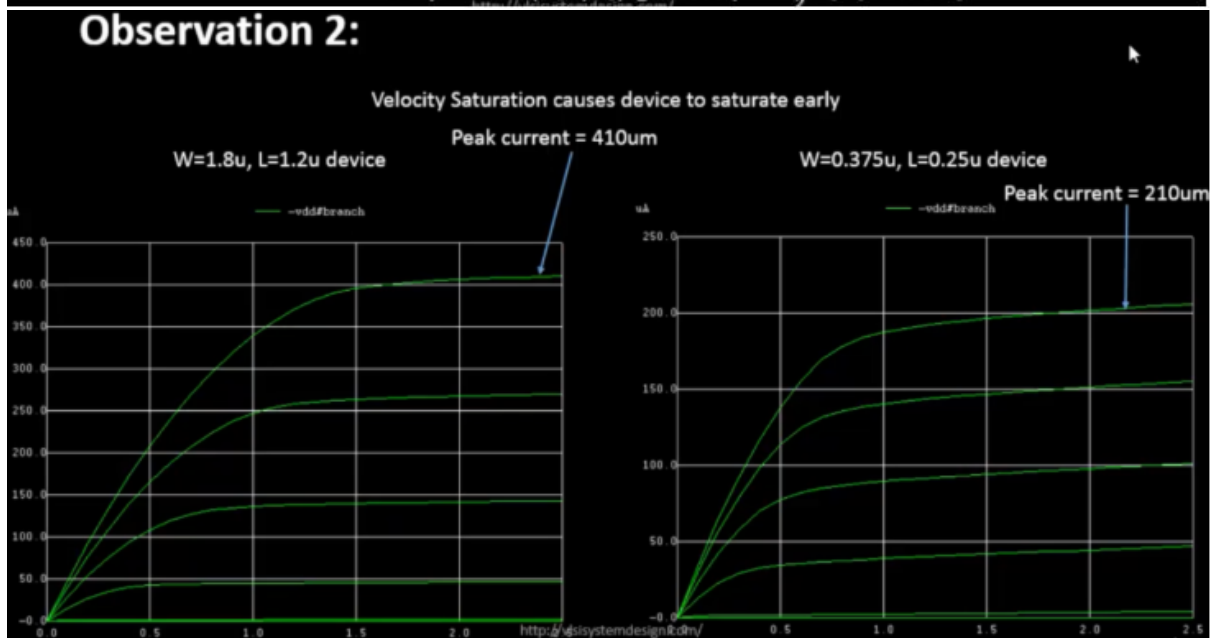
Where $V_{min} = \min(V_{gt}, V_{ds}, V_{dsat})$

For eg., If $V_{dsat} = \text{minimum value}$

$$I_d = k_n \cdot \left[(V_{gt} \cdot V_{dsat}) - \frac{V_{dsat}^2}{2} \right] \cdot [1 + \lambda V_{ds}]$$

$$I_d = \mu_n \cdot C_{ox} \cdot (W/L) \cdot \left[(V_{gt} \cdot V_{dsat}) - \frac{V_{dsat}^2}{2} \right] \cdot [1 + \lambda V_{ds}]$$

Looks like current should increase at lower nodes



OBSERVATIONS:

Observation 1: For short-channel devices, at **higher electric fields**, the device enters **velocity saturation**, causing **Ids to remain constant** as **Vds increases**. Here, **Ids** becomes a **linear function of Vgs**, unlike the **quadratic dependence** in long-channel devices.

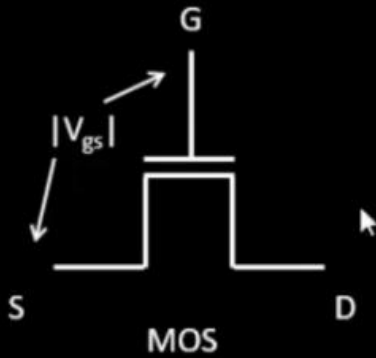
Observation 2: **Velocity saturation** causes the device to **saturate earlier**.

CMOS Voltage Transfer Characteristics (VTC):

- Vout is **high** when Vin is low, and Vout is **low** when Vin is high.
- The **transition region** shows a steep drop where both NMOS and PMOS conduct, defining the switching threshold.

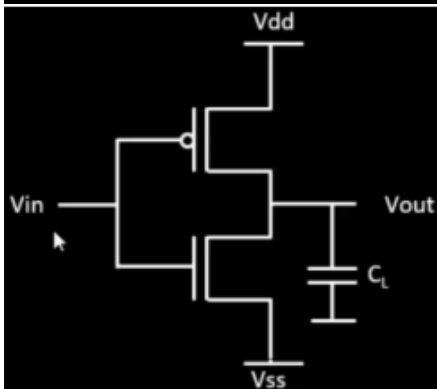
MOSFET as a switch

MOS device characteristics

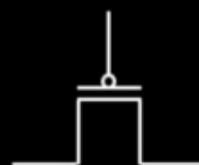


Transistor → Switch

- With infinite OFF resistance when $|V_{gs}| < |V_T|$
- With finite ON resistance when $|V_{gs}| > |V_T|$



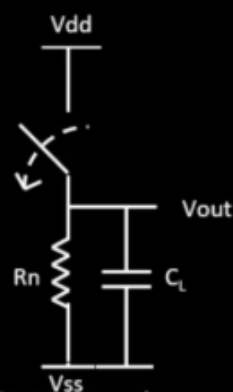
V_{in} is 'high' and equal to ' V_{dd} '



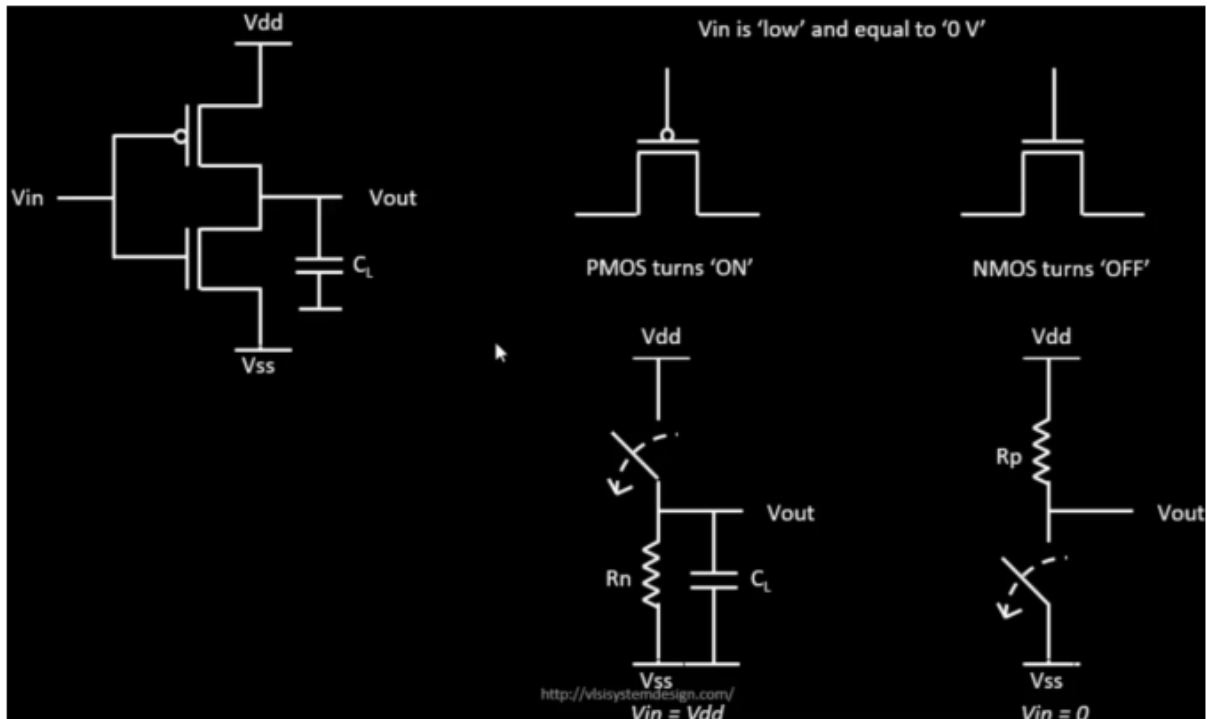
PMOS turns 'OFF'



NMOS turns 'ON'

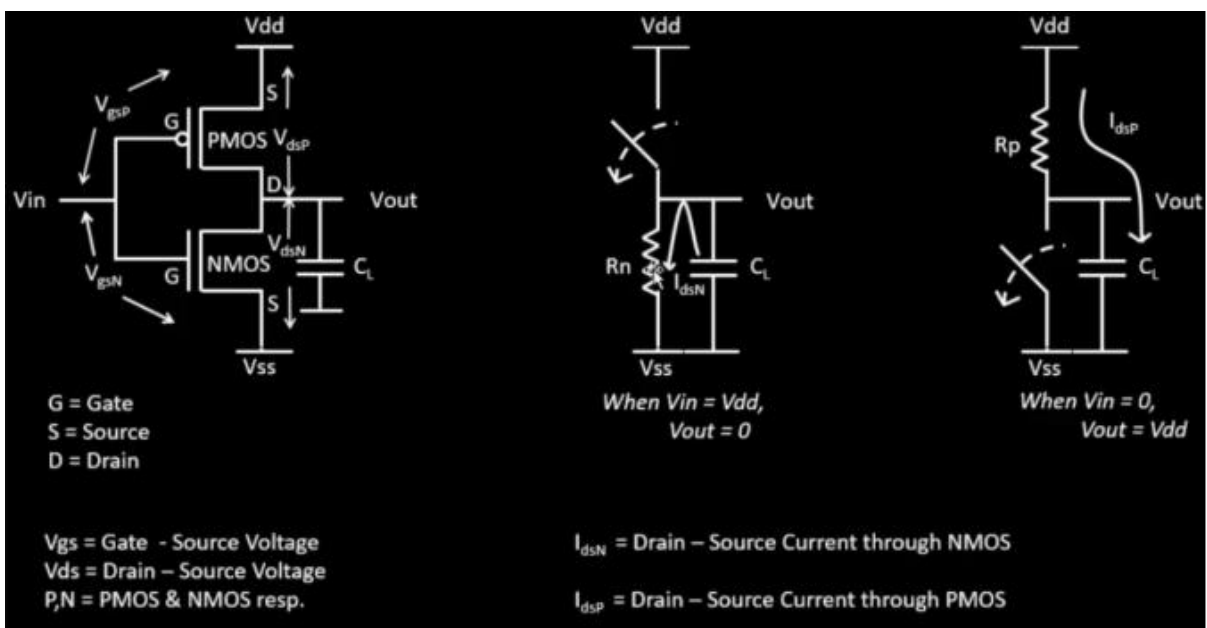


<http://vlsisystemdesign.com/>
 $V_{in} = V_{dd}$



****Introduction to Standard MOS Voltage-Current Parameters:****

In MOS devices, **R_p** (PMOS) and **R_n** (NMOS) act as **non-linear resistors**, where their resistance is a **function of drain current (I_{ds})**, influenced by gate voltage (V_{gs}) and drain voltage (V_{ds}).



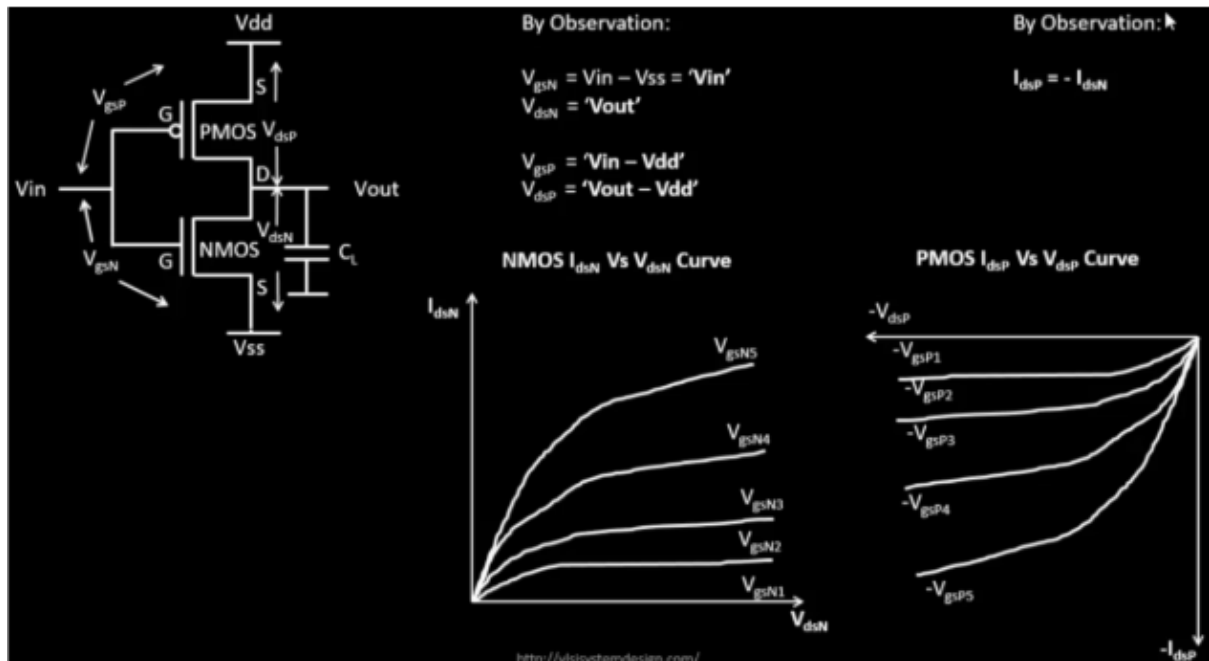
****PMOS/NMOS Drain Current vs. Drain Voltage:****

1. **Linear Region ($V_{ds} < V_{dsat}$):**

- Drain current (**I_{ds}**) increases **linearly** with **V_{ds}** .
- Device behaves like a **resistor** controlled by V_{gs} .

2. **Saturation Region ($V_{ds} \geq V_{dsat}$):**

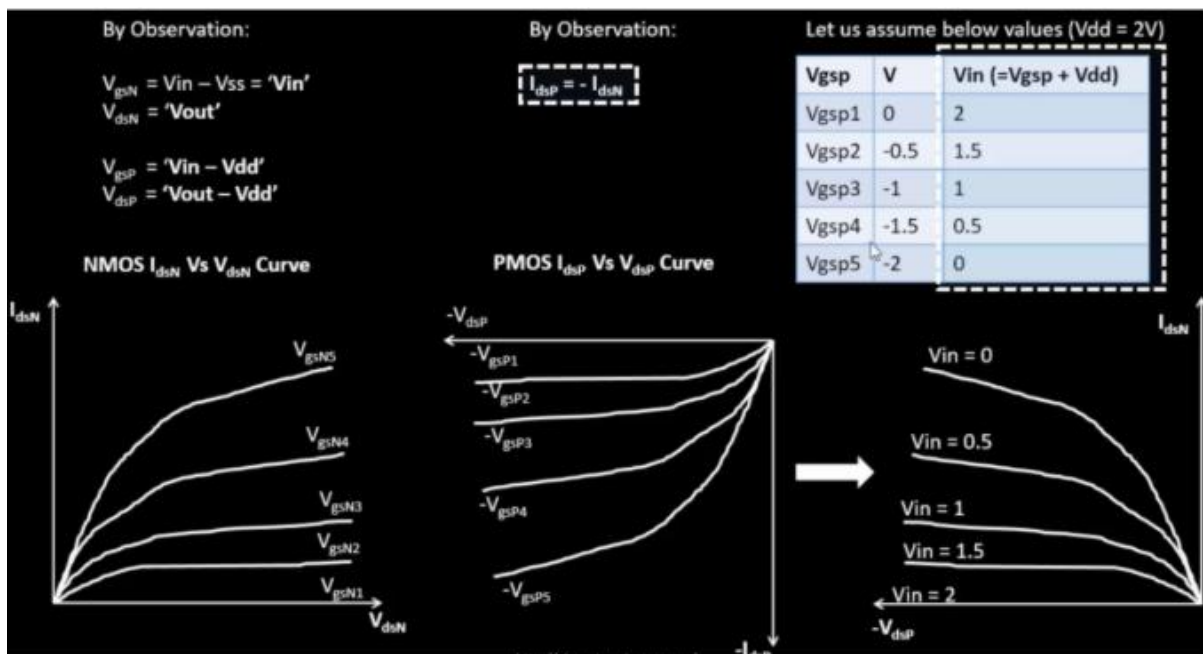
- Drain current (I_{ds}) becomes **constant** and independent of V_{ds} .
- For **NMOS**, I_{ds} depends on $V_{gs} - V_{th}$.
- For **PMOS**, I_{ds} depends on $V_{th} - V_{gs}$.



Step1. Convert **PMOS gate-source voltage** to **V_{in}** .

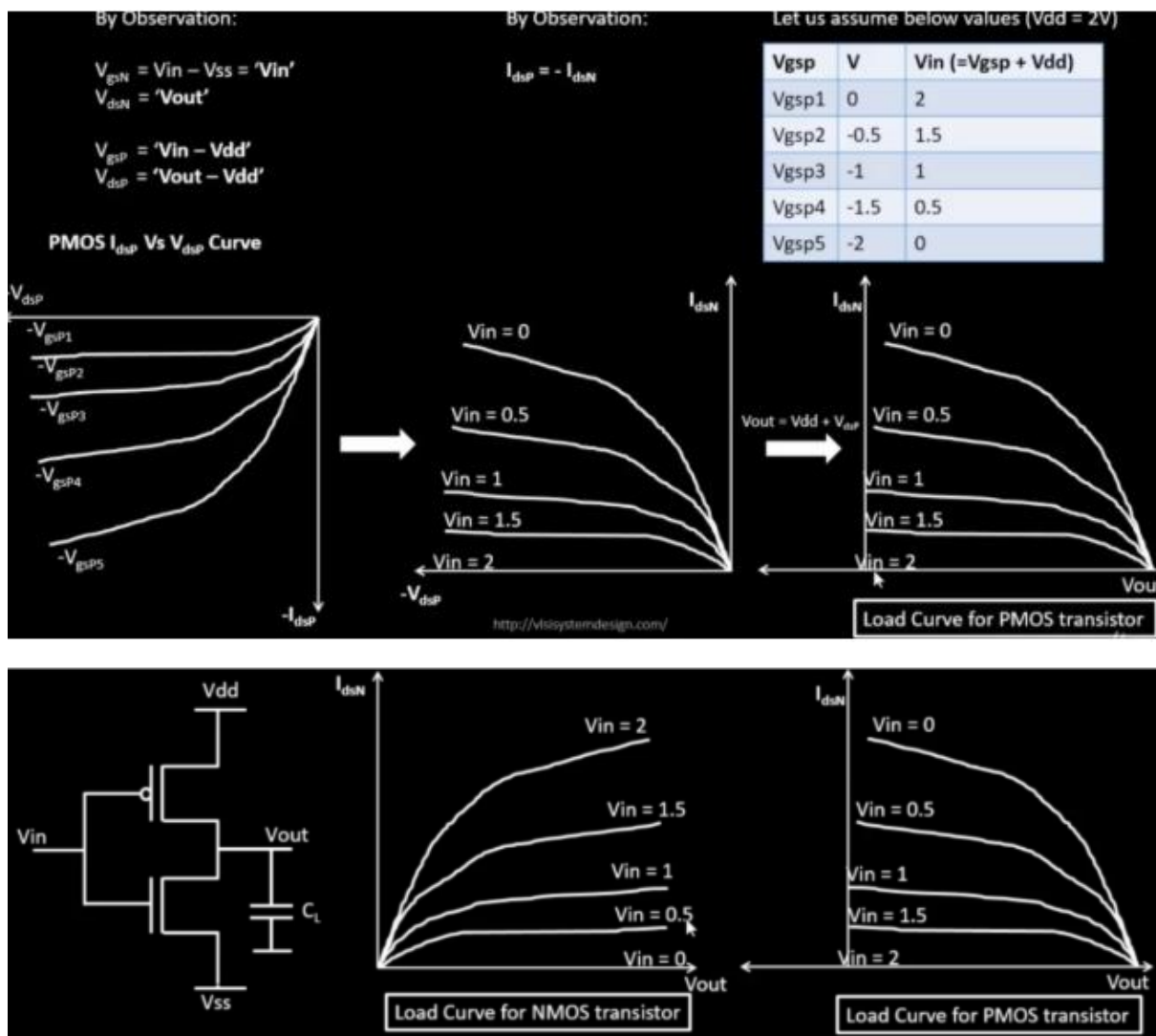
Replace internal node voltages with **V_{in}** , **V_{dd}** , **V_{ss}** , and **V_{out}** .

Simulate and plot the **VTC** by varying **V_{in}** and analyzing **V_{out}** .



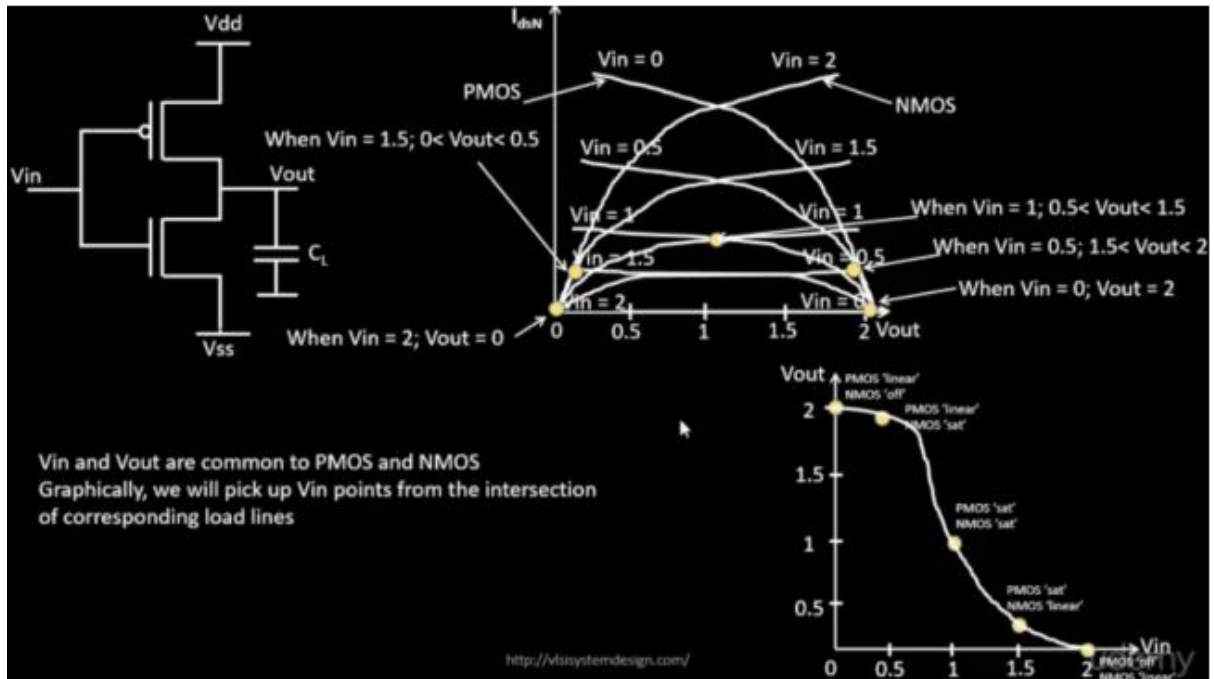
****Step 2 & Step 3:****

- Convert **PMOS** and **NMOS** drain-source voltages to **Vout**.
- Relate the drain-source voltages of both devices to the output voltage (**Vout**) by considering their respective operating regions (linear or saturation).



Step 4:

- **Merge the PMOS and NMOS load curves** by combining their drain current (I_{ds}) characteristics with respect to **Vout**.
- **Plot the Voltage Transfer Characteristic (VTC)** by mapping **Vout** against **Vin**, showing the transition from low to high output voltage.



)

LABS

SPICE simulation for lower nodes and velocity saturation effect

Sky130 Id-Vgs

EXAMPLE 1

*Model Description

.param temp=27

*Including sky130 library files

.lib "sky130_fd_pr/models/sky130.lib.spice" tt

*Netlist Description

XM1 Vdd n1 0 0 sky130_fd_pr__nfet_01v8 w=0.39 l=0.15

R1 n1 in 55

Vdd vdd 0 1.8V

Vin in 0 1.8V

*simulation commands

```
.op
```

```
.dc Vdd 0 1.8 0.1 Vin 0 1.8 0.2
```

```
.control
```

```
run
```

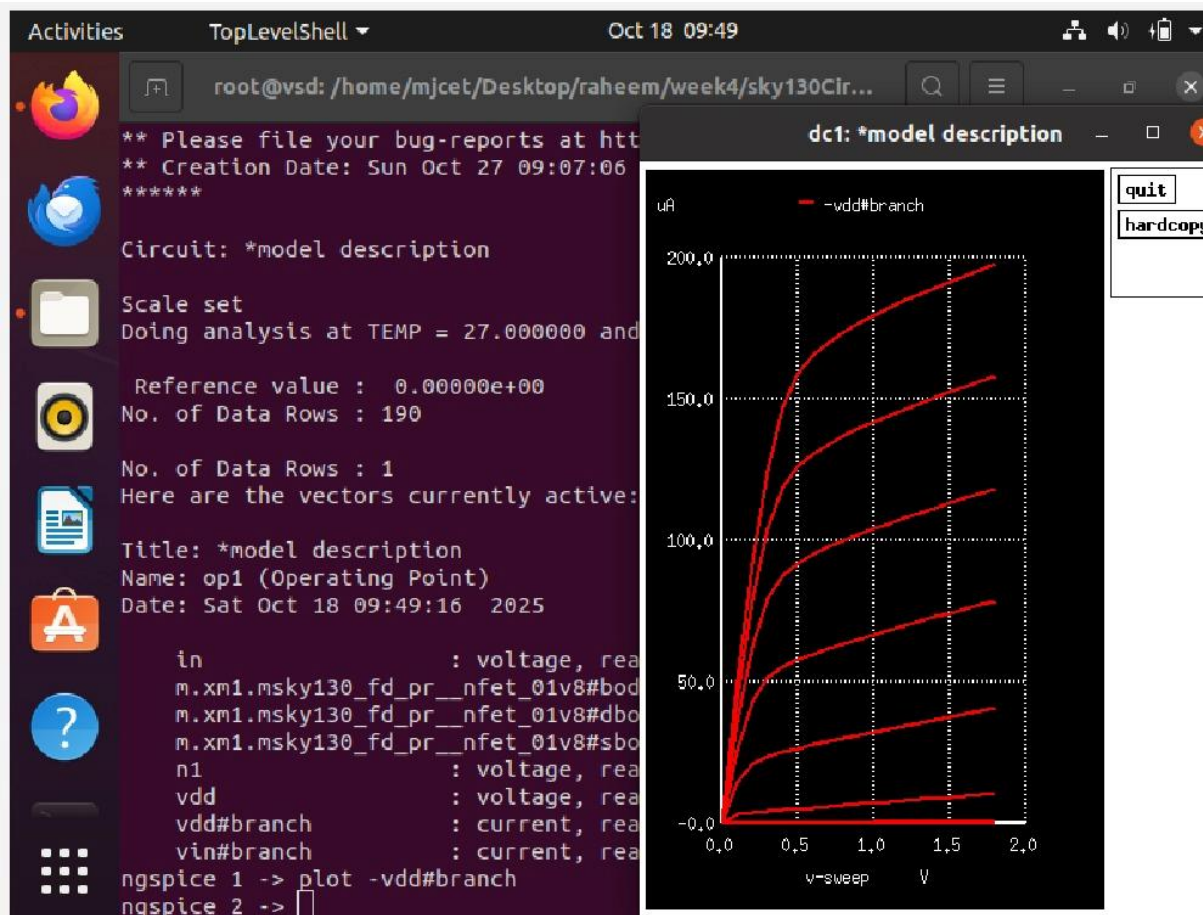
```
display
```

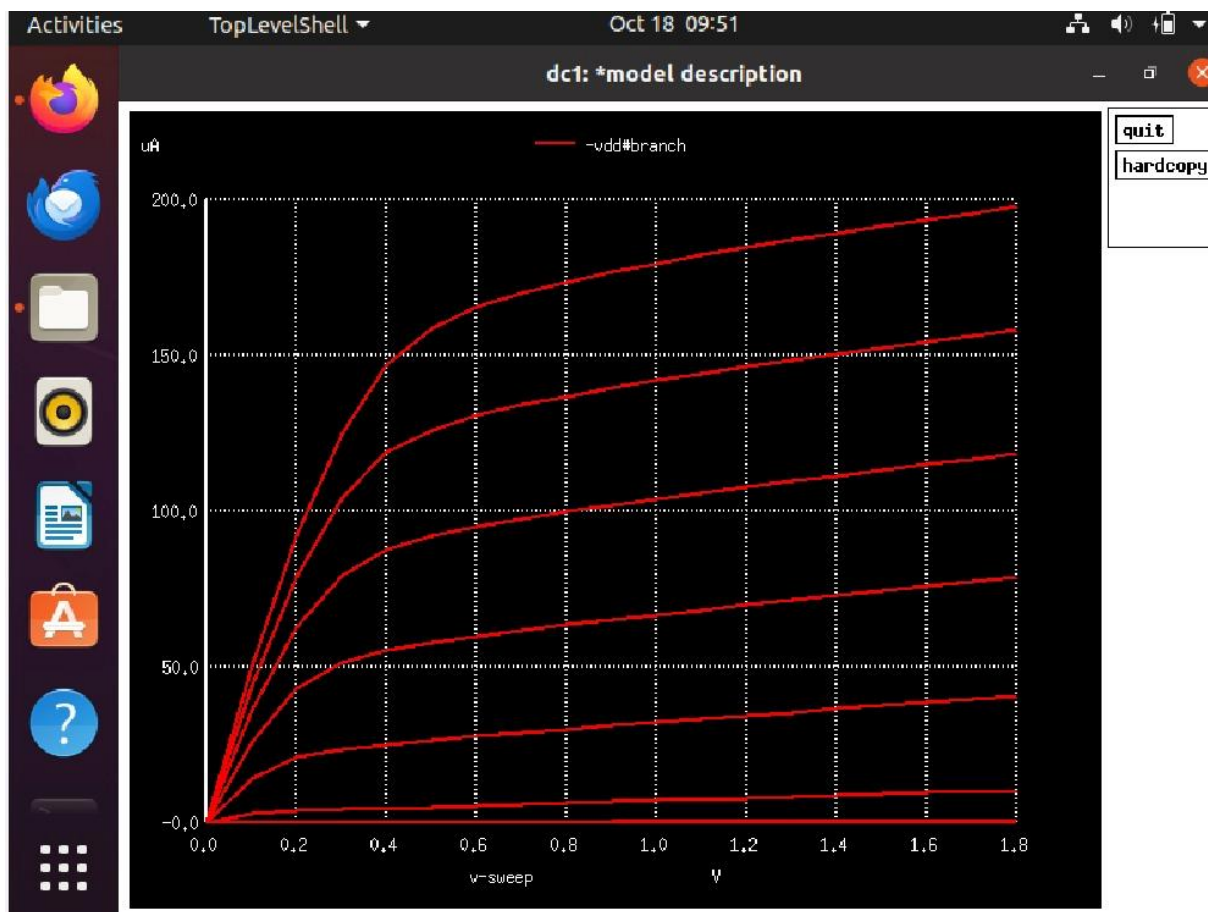
```
setplot dc1
```

```
.endc
```

```
.end
```

The plot of I_{ds} vs V_{ds} over constant V_{gs} :





EXAMPLE 2

*Model Description

```
.param temp=27
```

*Including sky130 library files

```
.lib "sky130_fd_pr/models/sky130.lib.spice" tt
```

*Netlist Description

```
XM1 Vdd n1 0 0 sky130_fd_pr__nfet_01v8 w=0.39 l=0.15
```

```
R1 n1 in 55
```

```
Vdd vdd 0 1.8V
```

```
Vin in 0 1.8V
```

*simulation commands

```
.op
```

```
.dc Vin 0 1.8 0.1
```

```
.control
```

```
run
```

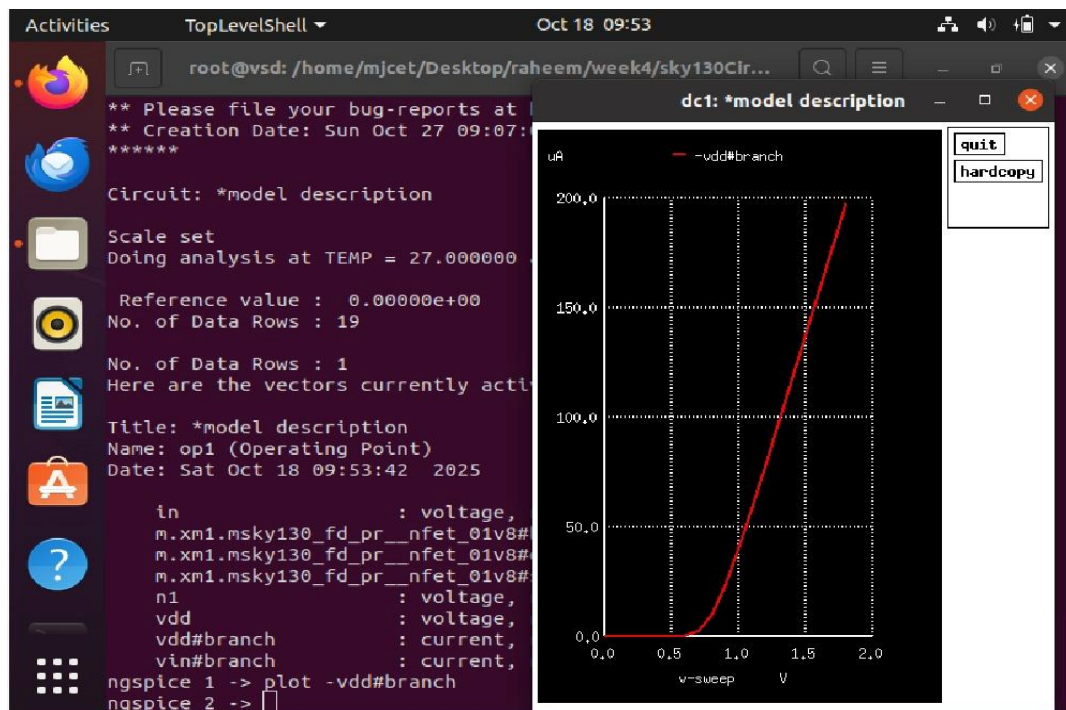
```
display
```

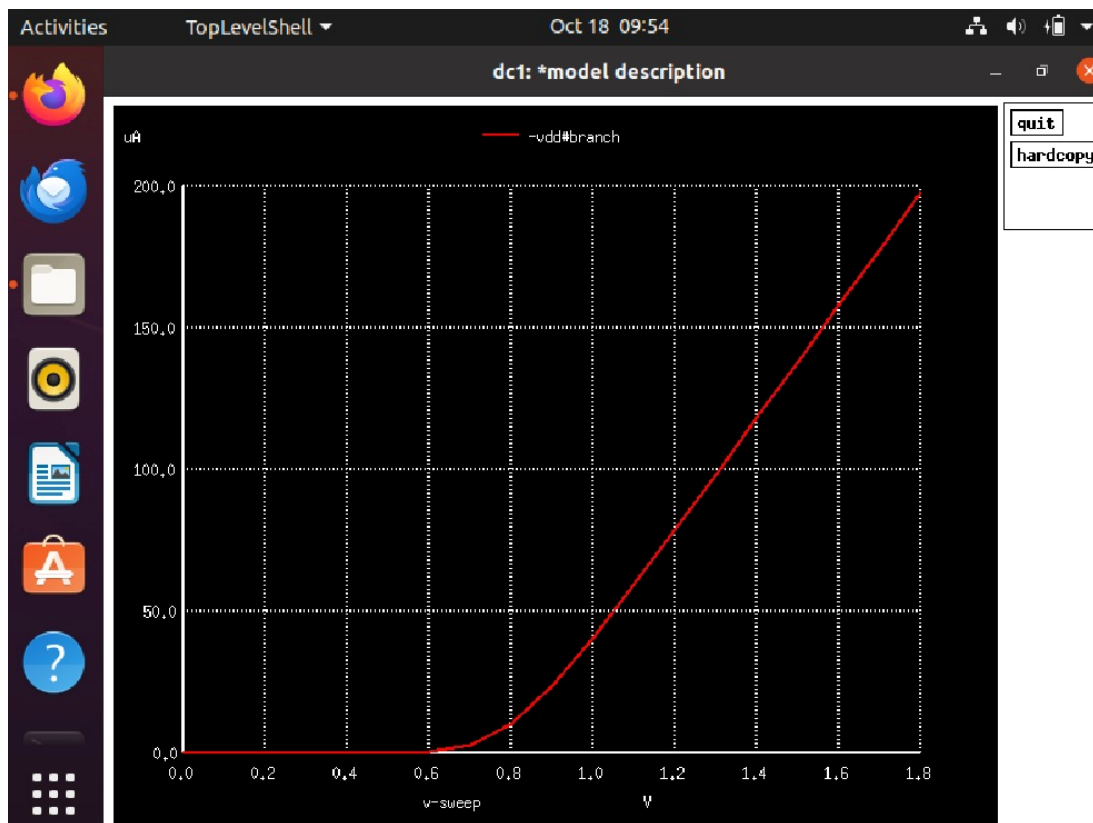
```
setplot dc1
```

```
.endc
```

```
.end
```

The plot of I_{ds} vs V_{gs} over constant V_{ds} :





Thank you VSD Team for mentoring the RSIC V Reference SoC Tapeout Program

Dr M A Raheem

Associate Professor Dept of ECE, MJCET, Hyderabad

abdulraheem@ieee.org

abdulraheem@mjcollege.ac.in

Orcid Id 0000-0001-8438-1477

Scopus ID: 55609453100

Web of Science ResearcherID AAD-7050-2021